

AP-E13 Annular Target-Valued Endpoint Replacement: A Wess–Zumino No-Go, a Scale-Compatible Repair, and the Reverse-Hölder Closure Test

Route-F derivation ledger

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Abstract

AP-E12 asked for a target-valued annular replacement which keeps the trace on a good sphere and makes all three squared complete-minor families equiintegrable. AP-E13 proves that this statement is false if “low shell energy” is its only hypothesis. For the geodesic-sphere trace $g_\varepsilon(\omega) = (\cos \varepsilon, \sin \varepsilon \omega)$, take the radially constant map on $A_{\rho, 2\rho}$ and set $\varepsilon = \sqrt{\rho}$. Its first-, second-, and third-minor shell norms are $O(\rho^2)$, $O(\rho)$, and zero. Nevertheless every smooth $\mathbb{S}^3$ -valued filling of the trace satisfies

$$\int_{B_\rho} |M_3(DU)|^2 dx \geq \frac{V(\varepsilon)^2}{|B_\rho|} \longrightarrow \frac{4\pi}{3}, \quad V(\varepsilon) = 2\pi \left(\varepsilon - \frac{1}{2} \sin 2\varepsilon \right).$$

The lower bound is forced by the Wess–Zumino volume modulo $\text{vol}(\mathbb{S}^3) = 2\pi^2$. Since every sphere in the shell has the same trace, selection of a good sphere cannot remove it. The corresponding third-minor densities concentrate on shrinking balls and are not equiintegrable.

A corrected positive theorem is also proved. If a good-sphere trace lies in a normal ball about q and $\|\log_q g\|_{L^\infty} \leq \Lambda\rho$, its geodesic contraction preserves the trace and controls all complete-minor norms by the shell norms. The linear-in- ρ oscillation is exactly the scale which compensates the cubic minor. Finally, the exact strong-quasiconvexity identity is tested as a possible independent reverse-Hölder route. Its highest cutoff-stress term leaves $\rho^{-2}|u - q|^2|M_2(Du)|^2$; without a Morrey oscillation estimate or prior higher integrability this is circular. Thus the universal annular lemma is disproved, the repaired conditional lemma is closed, but endpoint graph density, regularity of the selected relaxed minimizer, the Riemann Hessian, and determinant variation remain unauthorized. The reproducible scaling and quadrature card passes 9/9.

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1 The proposed endpoint lemma and its exact meaning

For $F \in \mathbb{R}^{4 \times 3}$ let $M_k(F)$ be the vector of all k -minors. The complete-minor part of the AP-E11 continuum action is

$$W(F) = \frac{1}{2} (|F|^2 + a_2 |M_2(F)|^2 + a_3 |M_3(F)|^2), \quad a_2, a_3 > 0. \quad (1)$$

Write $A_{\rho, 2\rho} = B_{2\rho} \setminus \overline{B_\rho}$ and

$$E_k(u; A_{\rho, 2\rho}) = \int_{A_{\rho, 2\rho}} |M_k(Du)|^2 dx, \quad M_1(Du) = Du. \quad (2)$$

The requested universal statement would have to imply at least the following vanishing property:

Definition 1.1 (Universal vanishing annular replacement). *Whenever $u_j : A_{\rho_j, 2\rho_j} \rightarrow \mathbb{S}^3$ and $\sum_{k=1}^3 E_k(u_j; A_{\rho_j, 2\rho_j}) \rightarrow 0$, one can select $s_j \in (\rho_j, 2\rho_j)$ and find $U_j : B_{s_j} \rightarrow \mathbb{S}^3$ with $\text{tr } U_j = u_j|_{S_{s_j}}$ such that*

$$\sum_{k=1}^3 \int_{B_{s_j}} |M_k(DU_j)|^2 dx \longrightarrow 0. \quad (3)$$

For endpoint graph density one needs more: after extension by zero to a fixed ambient ball, the three squared-minor families should be equiintegrable.

Ordinary Luckhaus interpolation controls a first-order Sobolev cost [1]. It does not contain the L^2 cubic-minor conclusion in (3). Trace extension for manifold-valued Sobolev maps can itself carry critical analytic and homotopy obstructions [2]. Here $\pi_2(\mathbb{S}^3) = 0$ removes the ordinary null-homotopy obstruction, but it does not remove the quantitative Wess–Zumino cost of a chosen null homotopy.

2 The geodesic-cap trace

Use polar coordinates on the unit three-sphere,

$$y = (\cos \theta, \sin \theta \omega), \quad 0 \leq \theta \leq \pi, \quad \omega \in \mathbb{S}^2, \quad (4)$$

for which

$$ds_{\mathbb{S}^3}^2 = d\theta^2 + \sin^2 \theta ds_{\mathbb{S}^2}^2, \quad \omega_3 = \sin^2 \theta d\theta \wedge \omega_2, \quad \int_{\mathbb{S}^3} \omega_3 = 2\pi^2. \quad (5)$$

For $0 < \varepsilon < \pi/2$ define

$$g_\varepsilon(\omega) = (\cos \varepsilon, \sin \varepsilon \omega). \quad (6)$$

Its two nonzero singular values on the unit $\mathbb{S}^2$ both equal $\sin \varepsilon$. Consequently

$$\int_{\mathbb{S}^2} |D_\omega g_\varepsilon|^2 d\omega = 8\pi \sin^2 \varepsilon, \quad \int_{\mathbb{S}^2} |M_2(D_\omega g_\varepsilon)|^2 d\omega = 4\pi \sin^4 \varepsilon. \quad (7)$$

The oriented volume of the cap bounded by g_ε is

$$\begin{aligned} V(\varepsilon) &= 4\pi \int_0^\varepsilon \sin^2 \theta d\theta = 2\pi \left(\varepsilon - \frac{1}{2} \sin 2\varepsilon \right) \\ &= \frac{4\pi}{3} \varepsilon^3 - \frac{4\pi}{15} \varepsilon^5 + O(\varepsilon^7). \end{aligned} \quad (8)$$

3 A Wess–Zumino lower bound for every filling

Lemma 3.1 (Relative degree quantizes the filling volume). *Let $U : B_\rho \rightarrow \mathbb{S}^3$ be smooth with trace $U|_{S_\rho} = g_\varepsilon$ after identifying $S_\rho/\rho \simeq \mathbb{S}^2$. Then some $k \in \mathbb{Z}$ satisfies*

$$\int_{B_\rho} U^* \omega_3 = V(\varepsilon) + 2\pi^2 k. \quad (9)$$

In particular, for sufficiently small ε ,

$$\left| \int_{B_\rho} U^* \omega_3 \right| \geq V(\varepsilon). \quad (10)$$

Proof. Glue U to the oppositely oriented canonical geodesic cap along their common trace. The result is a map $\mathbb{S}^3 \rightarrow \mathbb{S}^3$ of integer degree k . The integral of ω_3 over the glued domain is therefore $2\pi^2 k$, which gives (9) up to reversing k . Since $0 < V(\varepsilon) < \pi^2$ for small ε , the closest element of $V(\varepsilon) + 2\pi^2 \mathbb{Z}$ to zero is $V(\varepsilon)$. $\square$

For an $\mathbb{S}^3$ -valued map the three derivative columns are tangent to the target. Hence the scalar pullback Jacobian is the component of the Hodge dual third-minor vector along U , and

$$|U^* \omega_3| \leq |M_3(DU)| \, dx. \quad (11)$$

Cauchy–Schwarz and (10) now give the central estimate.

Theorem 3.2 (Wess–Zumino filling bound). *Every trace-preserving filling above satisfies*

$$\boxed{\int_{B_\rho} |M_3(DU)|^2 \, dx \geq \frac{V(\varepsilon)^2}{|B_\rho|} = \frac{3V(\varepsilon)^2}{4\pi\rho^3}.} \quad (12)$$

The conclusion extends to a complete-minor graph limit for which the trace and distributional Jacobian are stable.

Proof. Equations (10) and (11) imply

$$V(\varepsilon) \leq \int_{B_\rho} |M_3(DU)| \, dx \leq |B_\rho|^{1/2} \|M_3(DU)\|_{L^2(B_\rho)}.$$

Squaring proves (12). Stability under graph convergence follows by strong L^2 convergence of the third-minor vector and the trace. $\square$

4 Disproof of the unconditional annular replacement

On $A_{\rho,2\rho}$ set

$$u_{\rho,\varepsilon}(r\omega) = g_\varepsilon(\omega). \quad (13)$$

There are two tangential singular values $\sin \varepsilon/r$ and one zero radial singular value. Direct integration yields

$$\begin{aligned} E_1(u_{\rho,\varepsilon}; A_{\rho,2\rho}) &= 8\pi\rho \sin^2 \varepsilon, \\ E_2(u_{\rho,\varepsilon}; A_{\rho,2\rho}) &= \frac{2\pi}{\rho} \sin^4 \varepsilon, & E_3(u_{\rho,\varepsilon}; A_{\rho,2\rho}) &= 0. \end{aligned} \quad (14)$$

Theorem 4.1 (No universal endpoint replacement). *The assertion (3) is false. More strongly, low complete-minor energy on a shell does not imply equiintegrable L^2 third minors for any trace-preserving interior replacement.*

Proof. Let $\varepsilon = \rho^{1/2}$ and send $\rho \downarrow 0$. Equation (14) gives

$$E_1 = 8\pi\rho^2 + O(\rho^3), \quad E_2 = 2\pi\rho + O(\rho^2), \quad E_3 = 0. \quad (15)$$

Thus the entire shell graph energy tends to zero. Every sphere in the shell has exactly the trace $g_{\sqrt{\rho}}$, so no good-radius selection changes the boundary data. On a selected radius $s \in (\rho, 2\rho)$, the filling bound and (8) give

$$\int_{B_s} |M_3(DU)|^2 dx \geq \frac{3V(\sqrt{\rho})^2}{4\pi s^3} \geq \frac{1}{8} \left(\frac{4\pi}{3} + o(1) \right). \quad (16)$$

If the replacement is made at the inner radius $s = \rho$, the limit is exactly $4\pi/3$. In either case it does not tend to zero, contradicting (3). Extend $|M_3(DU)|^2$ by zero to a fixed ball. Its integral on the shrinking set $B_{2\rho}$ remains bounded below, so the family is not equiintegrable. $\square$

The full scale diagram is useful. Put $\varepsilon = \rho^\alpha$. Then

$$E_1 \asymp \rho^{1+2\alpha}, \quad E_2 \asymp \rho^{4\alpha-1}, \quad \inf_U \int_{B_\rho} |M_3(DU)|^2 dx \gtrsim \rho^{6\alpha-3}. \quad (17)$$

The shell energy vanishes already for $\alpha > 1/4$, whereas the forced filling cost vanishes only for $\alpha > 1/2$. The interval $1/4 < \alpha \leq 1/2$ is the previously missing endpoint obstruction window. This does not contradict AP-E12's radial Malý-dipole calculation: that calculation studied a different interior two-hole mechanism and required finite second-minor energy all the way to its singular core. The present map lives on a punctured shell and asks whether that core can be replaced.

Remark 4.2 (Necessary boundary datum). *For a general trace $g : S_\rho \rightarrow \mathbb{S}^3$, let $\mathcal{V}(g) \in \mathbb{R}/(2\pi^2\mathbb{Z})$ be its relative Wess–Zumino volume. A necessary condition for vanishing cubic-minor filling cost is*

$$\frac{\text{dist}(\mathcal{V}(g), 2\pi^2\mathbb{Z})^2}{\rho^3} \longrightarrow 0. \quad (18)$$

The shell norm alone does not imply (18).

5 A scale-compatible positive replacement theorem

The no-go identifies the missing hypothesis. The following sufficient condition is stronger than (18), but it controls all three minor orders simultaneously.

Theorem 5.1 (Conditional target-valued endpoint replacement). *Let $g : S_\rho \rightarrow \mathbb{S}^3$ have image in a normal ball about $q \in \mathbb{S}^3$ and suppose*

$$\|\log_q g\|_{L^\infty(S_\rho)} \leq \Lambda\rho, \quad \Lambda\rho < \frac{1}{4} \text{inj}(\mathbb{S}^3). \quad (19)$$

Put

$$T_1(g) = \int_{S_\rho} |D_\tau g|^2 dS, \quad T_2(g) = \int_{S_\rho} |M_2(D_\tau g)|^2 dS. \quad (20)$$

There is a trace-preserving $U : B_\rho \rightarrow \mathbb{S}^3$ and a constant C , uniform for $\Lambda\rho$ in a fixed normal ball, such that

$$\int_{B_\rho} |DU|^2 dx \leq C\{\rho T_1 + \Lambda^2 \rho^3\}, \quad (21)$$

$$\int_{B_\rho} |M_2(DU)|^2 dx \leq C\{\rho T_2 + \Lambda^2 \rho T_1\}, \quad (22)$$

$$\int_{B_\rho} |M_3(DU)|^2 dx \leq C\Lambda^2 \rho T_2. \quad (23)$$

Proof. Write $v(\omega) = \log_q g(\rho\omega)$ and define the geodesic cone

$$U(r\omega) = \exp_q \left(\frac{r}{\rho} v(\omega) \right), \quad 0 \leq r \leq \rho. \quad (24)$$

It is $\mathbb{S}^3$ -valued, equals q at the centre, and has trace g . On the fixed normal ball, $D \exp_q$ and its exterior powers are uniformly bounded and invertible. Since r/ρ cancels the $1/r$ in a physical angular derivative,

$$|D_r U| \leq C\Lambda, \quad qquad |D_\tau U(r, \omega)| \leq C|D_\tau g(\rho\omega)|. \quad (25)$$

Decomposing exterior products into radial–tangential and purely tangential parts gives

$$|M_2(DU)|^2 \leq C\{ |M_2(D_\tau g)|^2 + \Lambda^2 |D_\tau g|^2 \}, \quad (26)$$

$$|M_3(DU)|^2 \leq C\Lambda^2 |M_2(D_\tau g)|^2. \quad (27)$$

The volume-to-boundary factor in polar integration is $\int_0^\rho r^2 dr / \rho^2 = \rho/3$. Integrating (25)–(27) proves (21)–(23). $\square$

If a shell map satisfies the scale-compatible oscillation bound and $E_k = E_k(u; A_{\rho, 2\rho})$, coarea selects one radius $s \in (\rho, 2\rho)$ at which, after changing universal constants,

$$T_1(u|_{S_s}) \leq C \frac{E_1}{\rho}, \quad qquad T_2(u|_{S_s}) \leq C \frac{E_2}{\rho}. \quad (28)$$

The theorem then bounds the filling by $C(E_1 + E_2 + \Lambda^2 \rho^3)$ at every minor order. For the cap family, $\varepsilon = \Lambda\rho$ makes all three costs $O(\rho^3)$. Thus linear-in-radius oscillation is a clean sufficient repair. It is not a consequence of the assumed endpoint graph energy; the critical counterexample has oscillation $\sqrt{\rho}$.

6 Can exact strong quasiconvexity bypass the no-go?

For $G = F^T F$, differentiation of (1) gives the classical stress

$$P(F) = F + a_2 \{ (\operatorname{tr} G)F - FG \} + a_3 F \operatorname{cof} G. \quad (29)$$

The singular-value representation implies

$$|P(F)| \leq C\{|F| + |F||M_2(F)| + |M_2(F)||M_3(F)|\}. \quad (30)$$

AP-E12 proved the exact ambient strong-quasiconvexity identity. For an actual smooth constrained minimizer, a target-valued cutoff comparison or formal tangent first variation would therefore try to estimate the highest stress term by

$$\frac{|u - q|}{\rho} |M_2(Du)||M_3(Du)| \leq \eta |M_3(Du)|^2 + \frac{C_\eta}{\rho^2} |u - q|^2 |M_2(Du)|^2. \quad (31)$$

The first term is absorbable. The second is not controlled at the endpoint: $M_2(Du)^2$ is only in L^1 , while Poincaré control of $u - q$ is not L^∞ . Hölder would require higher integrability of $M_2(Du)$, which is the conclusion the reverse-Hölder argument is meant to prove. An L^∞ Morrey estimate $\|u - q\|_\infty \leq c\rho$ would close the scale, but it is precisely the additional hypothesis of the positive annular theorem.

Proposition 6.1 (Reverse-Hölder closure verdict). *The exact strong-quasiconvexity identity does not, by itself, provide a non-circular reverse-Hölder seed for the sphere-constrained relaxed minimizer. A successful model-specific proof must add at least one new ingredient which controls (31): for example a Wess–Zumino-flux decay plus Morrey oscillation estimate, a monotonicity or frequency formula, or a compensated Noether/Maurer–Cartan estimate.*

This proposition is a boundary statement, not a proof that every possible reverse-Hölder theorem is false. Minimality may exclude the cap pattern for the selected $B = 1$ solution, but that requires a new model-specific theorem. Moreover, the selected object is presently only a minimizer of the fixed-degree lower-semicontinuous relaxation. Until local fixed-trace recovery is available, inserting the naive stress (29) into its weak Euler–Lagrange equation would assume the desired conclusion.

The caution is consistent with the older Skyrme variational literature: finite Skyrme energy maps need not be continuous, and smooth approximation with uniformly bounded Skyrme energy was explicitly identified as an unresolved step [3]. Integer degree and existence can instead be established directly in the energy class [4]. Recent progress for the Faddeev–Skyrme Hopf map proves global minimality in a special homogeneous large-coupling problem, but does not supply the present $\mathbb{S}^3$ complete-minor endpoint replacement [5]. A second 2026 result characterizes $W^{1,2}$ Hopf lifts by exactness of the pulled-back area form and constructs smooth approximations preserving that base-map/gauge constraint [6]. It does not control L^2 second and third minors, but it suggests an actionable change of variables for the next reverse-Hölder attempt: decompose the $\mathbb{S}^3$ field into its $\mathbb{S}^2$ Hopf projection and vertical gauge one-form, then test whether (31) becomes an exact div–curl pairing.

7 Numerical verification and authorization ledger

The independent card

`route.f/code/verify_ap_e13_annular_replacement_reverse_holder.py`

checks the exact trace norms, stable cap-volume series, Wess–Zumino lower bound, explicit geodesic filling, and power-law slopes. It passes 9/9. At the smallest production radius $\rho = 10^{-8}$ with $\varepsilon = \sqrt{\rho}$, the shell norm is 6.283×10^{-8} whereas the filling lower bound is 4.188790188, converging to $4\pi/3$. For $\varepsilon = \rho$, all measured slopes equal three to numerical precision.

The updated decision is

$$\boxed{\begin{aligned} C_{\text{universal annular}} &= \text{false (disproved)}, & C_{\text{conditional linear oscillation}} &= \text{true}, \\ C_{\text{reverse Hölder for selected minimizer}} &= \text{false (open)}, \\ C_{D_{\text{end}}} = C_{\text{classicality}} = C_{\text{isolation}} = C_{\text{Hessian}} = C_{\text{det}} = C_{\text{portal}} &= \text{false}. \end{aligned}} \quad (32)$$

The no-go eliminates one proposed proof strategy, not the endpoint density theorem itself. The most efficient next target is now narrower: prove that the selected relaxed $B = 1$ minimizer has (18) and scale-compatible trace oscillation on good spheres, preferably from a monotonicity/frequency identity. In parallel, one may exploit the $\mathbb{S}^3 \simeq SU(2)$ Maurer–Cartan and target-rotation Noether structures to cancel the last term in (30); the Hopf base/gauge split above is the first concrete formulation to test. Only after either route yields local recovery, classicality, and continuum isolation is the same-action Riemann Hessian legal.

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